AMCM — Agent Memory Continuity Module

OOF™ Origin Open Foundation™

Independent Methodological Authority

OriginID: OOF-OID-MEM-AMCM-2026-06-08-0001
Architecture Ecosystem: Structured Reality Standards™
Architecture Family: Memory Governance Intelligence Architecture (MGIA™)
Operational Layer: Agent Memory Governance Layer
Governed Space: Agent Memory Continuity
Category: AI & Interpretation
Subcategory: Agent Memory Continuity Governance
Type: Agent Memory Integrity Module
Parent Standard: Agent Memory Integrity Standard (AMIS)
Version: 1.0
Status: Canonical · Open Module
Origin Date: 8 June 2026
Compatibility: OOF Methodology OS · Agent Memory Integrity Standard (AMIS) · Memory Integrity Standard (MIS) · Memory Continuity Integrity Module (MCIM) · Agent Cognition Integrity Standard (ACIS) · Cognitive Evolution Governance Standard (CEGS) · INTEGROS® — Integrity Standard
AI-Readable: Yes
Authority: OOF
Protection: MIP — Methodological Intellectual Property
Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Agent Memory Continuity Module (AMCM) defines the structural conditions under which memory utilized by an autonomous agent remains continuous, persistent, retrievable, reconstructable, and operationally valid across time, tasks, sessions, environments, and operational states.

AMCM governs agent memory continuity.

The module ensures that agents preserve meaningful continuity of context, experience, learned knowledge, objectives, and operational understanding rather than becoming fragmented across interactions or operational cycles.

A system satisfies AMCM only if:

- memory continuity remains preservable
- contextual persistence remains assessable
- operational history remains reconstructable
- continuity disruptions remain detectable
- memory persistence remains governable

- continuity remains operationally valid

An agent that repeatedly loses relevant context or operational history does not satisfy AMCM.

Module Operational Role

AMCM defines the continuity-governance layer of AMIS.

Its role is to preserve persistent memory across the operational lifecycle of an autonomous agent.

Module Operational Space

AMCM governs:

- agent memory continuity
- contextual persistence
- operational memory persistence
- continuity reconstruction
- session continuity
- long-term memory continuity
- continuity accountability
- continuity governance

The module applies wherever agents must preserve memory across time.

Module Function

The module applies wherever systems must preserve:

- continuous operational context
- persistent memory availability
- continuity of learned experience
- governance-valid memory persistence
- reconstructable operational history
- reliable long-term agent operation

Its function is to ensure that autonomous agents remain capable of maintaining meaningful continuity of memory.

Minimum Implementation Framework

1. Define the Agent Memory Continuity Object

The organization must define which memory assets require continuity governance.
This may include:

- operational context
- learned knowledge
- user interactions
- objectives
- task history
- planning history
- environmental understanding
- agent experience

2. Define Agent Memory Continuity Conditions

The system must define the conditions under which continuity remains valid.
This includes:

- persistence requirements
- continuity requirements
- retrieval requirements
- reconstruction requirements
- context-preservation requirements
- governance-valid continuity conditions

3. Define Continuity Degradation Detection Logic

The system must define how continuity degradation is identified.
This may include:

- context loss
- memory fragmentation
- persistence failures
- broken operational history
- discontinuous learning
- continuity drift

4. Define Operational Response or Governance Logic

The system must define governance logic for continuity failures.
Governance response may include:

- continuity review
- memory restoration
- reconstruction procedures
- governance intervention
- escalation
- continuity protection
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- continuity states
- continuity disruptions
- restoration actions
- governance reviews
- intervention activities
- resulting memory states

An agent memory environment must not remain continuity-valid if materially significant memory continuity cannot be reconstructed, reviewed, or governed.

Use Case 1 — Persistent AI Assistant

Scenario

An AI assistant continuously supports a user across months or years of interaction while preserving relevant context, preferences, and historical understanding.

Application

AMCM governs continuity of memory, preservation of context, and long-term operational persistence.

Result

The assistant maintains stable continuity of interaction and avoids repeated loss of relevant knowledge.

Use Case 2 — Autonomous Industrial Agent

Scenario

An industrial agent continuously operates within a production environment while accumulating operational experience and historical knowledge.

Application

AMCM governs continuity of operational memory and preservation of accumulated experience across long-term operation.

Result

The agent gains stronger operational reliability, improved contextual awareness, and reduced exposure to memory fragmentation.

Canonical Closing Statement

Agent Memory Continuity Module (AMCM) defines the structural conditions under which memory utilized by an autonomous agent remains continuous, persistent, retrievable, reconstructable, and operationally valid across time, tasks, sessions, environments, and operational states.

Autonomous agents cannot maintain reliable operation if memory continuity collapses. Agent memory continuity therefore becomes a foundational integrity condition of trustworthy autonomous memory systems.

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

https://originopenfoundation.org/